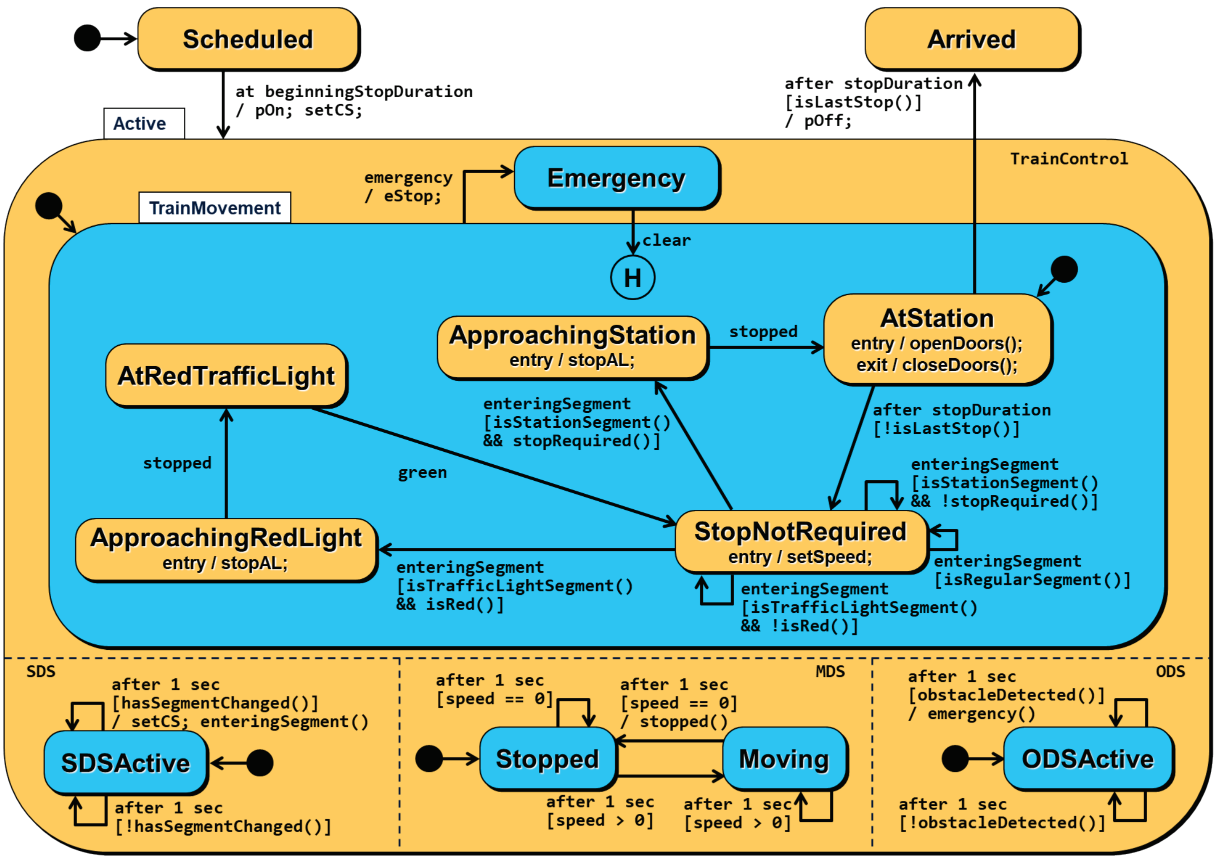
State Machine Question (100 marks)

Given the problem description below, specify the ***state machine for a train*** in the Advanced Train Automation System (ATAS). **Show the states as well as transitions including events, guards, and actions. Define composite states and orthogonal regions when needed to reduce complexity.** You do **not** need to specify parameters for events, guards, and actions. Use **meaningful** names for the private methods for guards and actions in your state machine. You do **not** need to define the method bodies of those methods.



An alternative solution without orthogonal regions replaces (i) the "emergency" event of the transition from TrainMovement to Emergency with "obstacleDetected", (ii) the "stopped" event of the transitions from ApproachingRedLight to AtRedTrafficLight and from ApproachingStation to AtStation with "stopDetected", and (iii) the "enteringSegment" event of the five transitions from StopNotRequired with "segmentChangeDetected". The setCS action would need to be added to all five transitions. The Active composite state is then not needed and the transition from Scheduled goes directly to TrainMovement. However, this solution does not capture at all that the SDS/MDS/ODS checks are performed every second.